M.N. NAFEEL

MECHANICAL ENGINEERING TECHNOLOGY GRADUATE |

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PROFESSIONAL SUMMARY

Mechanical Engineering Technology graduate with a strong foundation in engineering principles, mechanical systems, manufacturing processes, workshop practices, and product design and development. Strong analytical and problem-solving skills with a commitment to innovation, continuous improvement, and professional growth. Eager to apply engineering knowledge and contribute effectively to multidisciplinary teams in achieving organizational goals and engineering excellence.

SKILLS AND KNOWLEDGE

Software: SolidWorks | Inventor | AutoCAD | Fusion 360 | Ansys | MATLAB

Design & Analysis: Parametric Modelling | Machine Design | FEA | Mechanical System Design | Reverse Engineering

Manufacturing & Maintenance: Lean Manufacturing | TPM | Kaizen | Process Improvement | Root Cause Analysis | Preventive Maintenance | Workshop practices | Welding & Fabrication | Machining Operation

Workshop: Machining | Welding and Fabrication | OHS Practices

Engineering Theories: Thermodynamics | Material Science | Fluid Dynamics | Engineering Mechanics | Engineering Graphics

Programming: Python, C++

PROFESSIONAL EXPERIENCE

- **MANUFACTURING INTERN** **Nov 2025 - Apr 2026**

  Hemas Consumer Brands | Dankotuwa, Sri Lanka

- Supported daily FMCG manufacturing operations and coordinated with production, quality, maintenance, and engineering teams to meet safety, quality, and productivity requirements.
- Monitored production KPIs and supported OEE, downtime, and idle-time calculations to identify process losses and improvement opportunities.
- Contributed to process-optimization and floor-layout activities for internal and third-party manufacturing facilities, including filling, checkweighing, packing, and pallet-flow arrangements using Microsoft Visio.
- Supported improvement discussions related to auger-filling precision, packing operations, maintenance coordination, utilities, and operational efficiency.
- Participated in Lean Manufacturing, TPM, Kaizen, and 5S activities and assisted with the identification of manufacturing losses and corrective actions.

- **INDUSTRIAL TRAINEE** **Sep 2022 - Mar 2023**

  Sri Lanka Ports Authority | Colombo, Sri Lanka

- Supported preventive and corrective maintenance of mechanical, electrical, and electromechanical equipment to improve availability and operational reliability.
- Performed workshop operations including turning, milling, drilling, fitting, metal-arc welding, and fabrication under technical supervision.
- Assisted troubleshooting, servicing, and rewinding activities for single-phase and three-phase electric motors while following workplace safety procedures.

SELECTED ENGINEERING PROJECTS

- **Feasibility Analysis of a Small-Scale Plastic Pyrolysis System | 2025 - 2026**

  Final-Year Engineering Project

- Designed a conceptual 30 L batch pyrolysis reactor system with an external heating arrangement, vapor piping, and shell-and-tube condenser for small-scale waste-to-energy application.
- Evaluated material selection, operating temperature, heat demand, feedstock performance, projected oil yield, process safety, and system requirements using literature and secondary data.
- Completed CAPEX, OPEX, profitability, and payback analyses to assess technical and economic feasibility for application in Sri Lanka.

- **Green Maintenance Practices in Automotive Workshops | 2025**
  Academic Project
  - Analyzed environmentally responsible maintenance practices and proposed measures to improve workshop efficiency while reducing waste, energy use, and environmental impact.

MECHANICAL DESIGN PORTFOLIO

Independent Mechanical CAD Projects | 2026 - Present
Personal Engineering Portfolio | Kandy, Sri Lanka

- Designs and develops mechanical components and assemblies, including gearboxes, hydraulic cylinders, bearing housings, slider-crank mechanisms, crankshafts, and exhaust-nozzle concepts, based on functional, mechanical, and manufacturing requirements.
- Applies Design for X (DfX) principles, including Design for Manufacturing, Design for Assembly, and Design for Ergonomics, to improve manufacturability, usability, and overall product performance.
- Applies engineering design principles, parametric modelling, design intent, geometric dimensioning and tolerancing (GD&T), assembly constraints, motion analysis, sheet-metal design, and surface modelling throughout the design-development process.
- Prepares detailed 3D models, assembly layouts, exploded views, and engineering drawings to communicate design specifications and support manufacturing and assembly activities.
- Evaluates component fit, motion, interference, functionality, material suitability, and manufacturability to refine designs and identify potential design issues.
- Publishes selected design projects and supporting technical documentation through a personal engineering portfolio to demonstrate continued development in mechanical product design and engineering.

EDUCATION

Bachelors of (Honors) in Mechanical Engineering Technology **2024 - 2026**
Lincoln University College | Second Class Upper Division

City & Guilds Level 5 IVQ Advanced Technician Diploma in Mechanical Engineering ***2023 - 2025***
City & Guilds of London Institute

Diploma in Production Technology ***2020 - 2023***
Sri Lanka German Training Institute | Merit Pass

CERTIFICATIONS & PROFESSIONAL AFFILIATIONS

Certifications: Occupational Safety and Health | Chemical Handling Health & Safety | Firefighting & Rescue Training | Machine Learning for Engineers

Affiliations: Affiliate Student Member, Institution of Mechanical Engineers (IMechE) | Member, International Association of Engineers (IAENG)

LANGUAGES

Tamil (Native) | English (Working Proficiency) | Sinhala (Conversational)

REFERENCE

Available upon request